

LINE GAMES:

Reduces
churn,
improves
game
experience,
and creates
revenue
opportunities
with BigQuery

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

LINE GAMES

About LINE GAMES

Established by LINE Corp., operator of the multinational LINE global messaging service, LINE GAMES acquired mobile game publisher Nextfloor in 2018 to position itself as a leading game developer and publisher.

The business is expanding through global take-up of popular online games such as *Destiny Child* – part of a roster of casual, puzzle and role-playing mobile games. In addition, LINE GAMES is developing PC and console products ranging from MMORPG (Massively Multiplayer Online Role-Playing Games) and Survival MOBA

With BigQuery and Holistics, LINE GAMES reduced user churn, identified opportunities to improve the game experience, while

costs by up to 50
through moving f
internet data cen
cloud-based solu

(Multiplayer Online Battle Arena) to TPS RPG
(Third-Person Shooter Role Playing Games) to
fulfill its mission of creating games across all
platforms and genres.

Industries: Gaming

Location: Korea

Google Cloud Results

- Lowers costs
up to 50%
- Gains real-
time
visualization
of
performance
against KPIs
- Improves
planning and
revenue
opportunities
by
identifying
and targeting
user cohorts

Reduc
churn
by ide
opport
impro
exper



About Holistics

Founded by a team of engineers and business intelligence professionals, Holistics provides a flexible cloud data platform for automating business intelligence reporting and data engineering operations. The organization helps data teams streamline data preparation and reporting processes to obtain quick answers to questions. Holistics supports operations in 20 countries, processes more than 3 million queries run by data teams, and delivers an average response time of five hours.

For LINE GAMES (<https://line.games>)
fun experience in the environme
their choice is the difference bet
failure. Established by LINE Corp
multinational LINE global messag

operations. The organization helps data

Google Cloud (<https://cloud.google.com/>) and

BigQuery (<https://cloud.google.com/bigquery/>)

answers to questions. Holistics supports
operations in 20 countries, processes
more than 3 million queries run by data

LINE GAMES acquired mobile game development teams, and delivers an average response time of five hours. In 2018, LINE GAMES positioned itself as a leading mobile game developer and publisher.

The business is expanding through global take-up of popular online games such as *Destiny Child* – part of a roster of casual, puzzle and role-playing mobile games. In addition, LINE GAMES is developing PC and console games ranging from MMORPG (Massively Multiplayer Online Role-Playing Games) and Survival MOBA (Multiplayer Online Battle Arena) to TPS RPG (Third-Person Shooter Role Playing Games) to fulfill its mission of creating games across all platforms and genres.

"Mobile games had dominated the games marketplace over the last two years," says Ji-Eun Kim, Data Analyst, LINE GAMES. "However, gamers are moving 'beyond the platform' – just look at the success of multi-platform games such as *Fortnite* and PlayerUnknown's *Battlegrounds*. Ultimately, fun is the key element and we are determined to make our games attractive and interesting."

"We found
BigQuery far
superior to
existing
relational

databases in processing large volumes of data quickly. In addition, BigQuery supports standard SQL, so our team members did not need any further education about how to use the service."

—Ji-Eun Kim, Data Analyst, LINE GAMES

Understanding user preferences

Achieving this objective means understanding user behaviors and preferences in as close to real-time as possible and adapting games accordingly. The business had initially run servers, storage, and networking in an internet data center, but found this infrastructure poorly equipped to meet its data capture, storage, and analysis needs. "We wanted to create a data lake, comprising of data in its raw state," says Dae-Kaun Kang, Data Engineer, LINE GAMES. "This would support a system to capture

and analyze structured and unstructured data, such as information from logs and users' gaming activities."

LINE GAMES' existing business intelligence solution, which relied on a traditional spreadsheet and inputs from one data analyst and one engineer, could not support the organization's growing need to identify indicators and figures on the internet. LINE GAMES wanted to build an Online Transaction Processing (OLTP) system that would provide a real-time view of changes that affect performance against key performance indicators and an Online Analytical Processing (OLAP) system to analyze and reduce user churn – the percentage of users that stop playing its games.

This project would, according to Kang, enhance LINE GAMES' use of data to maintain market leadership and operate more efficiently. "We use data analysis for critical activities such as marketing and game management," says Kang. "For example, we decide whether to continue marketing campaigns for certain games by comparing the lifetime value of users with user acquisition costs, enabling us to measure how long it takes to recover our investment in each user."

"Furthermore, while the cloud is reducing the ability of businesses to gain an edge by building information systems ahead of their competitors, the effective management and utilization of big data still presents an opportunity to secure market leadership," adds Kang.

A comprehensive market evaluation

LINE GAMES conducted a comprehensive market analysis of systems that could meet its OLAP and OLTP requirements. "We needed a product that could do this for us and we chose a cloud-based business intelligence and data analytics platform from Holistics (<https://www.holistics.io/>)," says Jae-Hwan Ryu, Data Engineer, LINE GAMES. "This system effectively utilizes the BigQuery (<https://cloud.google.com/bigquery/>) serverless, scalable cloud data warehouse."

"With Holistics and BigQuery, we can see in real-time how each of our projects is performing against key performance indicators and visualize in multiple dimensions the

data held in the
data
warehouse."

—Seong-In Kim, Data
Analyst & Team Leader,
LINE GAMES

"We found BigQuery far superior to existing relational databases in processing large volumes of data quickly," adds Ryu. "In addition, BigQuery supports standard SQL, so our team members did not need any further education about how to use the service."

While LINE GAMES considered a range of data visualization and business intelligence products, Holistics' user interface and user experience, as well as its filter and support for cube functions, proved compelling. The Holistics filter function enables users to customize reports and modify reports views, while each dimension defined by the cube structure is easily accessible to general users as well as data scientists, enabling team members to check measurement values for data across various user cohorts with just a few clicks.

Up to 50 percent cost reduction

The business migrated its data to a system based on BigQuery and Holistics over a two-week period and reaped immediate rewards. The most

immediate benefit was a cost reduction of up to 50 percent due to moving from an internet data center to a cloud-based data warehousing and analysis solution.

However, the solution's longer term value lies in enabling LINE GAMES to use data to improve the gaming experience, operate more efficiently, and seize opportunities to increase revenues and enter new markets. "With Holistics and BigQuery, we can see in real-time how each of our projects is performing against key performance indicators and visualize in multiple dimensions the data held in the data warehouse," says Isaac Park, Data Analyst, LINE GAMES.

The system is now well established and used by multiple teams within LINE GAMES. The business' game planning team can slice and dice users by cohort and identify their behaviors for planning purposes, while the marketing team can identify activity flows and return on ad spend by channel.

"BigQuery is
delivering on our
requirement for
fast
performance
with large

volumes of data,
enabling us to
make better
business
decisions faster."

—Isaac Park, Data
Analyst, LINE GAMES

This team can compare customer lifetime value with user acquisition costs to determine which campaigns should be pursued and the channels they are most effective on. The publishing team can identify which in-app purchase products are efficient and selling well to specific user cohorts.

"With Holistics and BigQuery, we are achieving a range of efficiencies," explains Kim. "For example, the Holistics filter and Explore Result functions enable us to view user flows – the directions taken to complete a task within our games – easily. We also use the email schedule feature to automate weekly reports previously completed manually, reducing our working time."

"BigQuery is delivering on our requirement for fast performance with large volumes of data, enabling us to make better business decisions faster," adds Kim.

User churn rates falling

Furthermore, user churn rates have fallen by up to 5 percent and user lifetime value has extended as LINE GAMES identifies and manages issues within games. "If you identify the variables that impact on user churn, you can quickly implement remediation measures such as giving affected users gifts or promotional benefits," says Kim.

This insight enabled LINE GAMES to minimize the impact of a minor issue that affected the performance of its signature game in Korea. "The launch of the game was a huge success," explains Kim. "However, we could see a cohort of newer users was leaving the game at a certain point. It turned out that in a player-versus-player environment, that cohort was experiencing a lower than anticipated winning rate."

"Our algorithm was matching these newer users with more experienced users, so they were losing frequently, becoming annoyed and leaving the game. Our development team fixed the algorithm, and we saw our retention rates and revenue increase."

With BigQuery and Holistics, LINE GAMES is well-positioned to continue using data to obtain a competitive advantage and expand into new markets and user communities.

Learn More

Reach out to
our team to
see how
Google Cloud
can help your

business.